

EXPERIMENT 4

AIM:

Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN.

SQL Queries

1. INNER JOIN

Display employees along with their department names.

SELECT

**e.Emp_ID,
e.Emp_Name,
e.Salary,
d.Dept_Name**

FROM Employee e

INNER JOIN Department d

ON e.Dept_ID = d.Dept_ID;

Emp_ID	Emp_Name	Salary	Dept_Name
1	Aarav Sharma	55000.00	IT
2	Riya Gupta	62000.00	IT
3	Arjun Verma	48000.00	IT
4	Ananya Singh	70000.00	IT
5	Karan Mehta	45000.00	IT
6	Priya Sharma	58000.00	IT
7	Neha Kapoor	50000.00	HR
8	Rahul Jain	65000.00	HR
9	Simran Kaur	42000.00	HR
10	Aditya Roy	46000.00	HR
11	Megha Joshi	55000.00	HR
12	Vivek Malhotra	72000.00	HR
13	Ishita Rao	60000.00	Finance
14	Manish Kumar	75000.00	Finance
15	Pooja Agarwal	52000.00	Finance
16	Rohit Bansal	68000.00	Finance
17	Kavya Nair	58000.00	Finance
18	Sahil Gupta	80000.00	Finance
19	Tanya Singh	47000.00	Marketing
20	Yash Verma	56000.00	Marketing
21	Nisha Patel	62000.00	Marketing
22	Aman Khan	44000.00	Marketing
23	Shreya Das	51000.00	Marketing
24	Dev Sharma	59000.00	Marketing
25	Akash Yadav	49000.00	Operations
26	Sneha Roy	57000.00	Operations
27	Varun Singh	63000.00	Operations
28	Komal Gupta	46000.00	Operations
29	Nitin Sharma	54000.00	Operations
30	Aditi Mehta	69000.00	Operations

2. LEFT JOIN

Display all departments and their employees.

SELECT

d.Dept_ID, d.Dept_Name, e.Emp_Name

FROM Department d

LEFT JOIN Employee e

ON d.Dept_ID = e.Dept_ID;

Dept_ID	Dept_Name	Emp_Name
1	IT	Aarav Sharma
1	IT	Riya Gupta
1	IT	Arjun Verma
1	IT	Ananya Singh
1	IT	Karan Mehta
1	IT	Priya Sharma
2	HR	Neha Kapoor
2	HR	Rahul Jain
2	HR	Simran Kaur
2	HR	Aditya Roy
2	HR	Megha Joshi
2	HR	Vivek Malhotra
3	Finance	Ishita Rao
3	Finance	Manish Kumar
3	Finance	Pooja Agarwal
3	Finance	Rohit Bansal
3	Finance	Kavya Nair
3	Finance	Sahil Gupta
4	Marketing	Tanya Singh
4	Marketing	Yash Verma
4	Marketing	Nisha Patel
4	Marketing	Aman Khan
4	Marketing	Shreya Das
4	Marketing	Dev Sharma
5	Operations	Aakash Yadav
5	Operations	Sneha Roy
5	Operations	Varun Singh
5	Operations	Komal Gupta
5	Operations	Nitin Sharma
5	Operations	Aditi Mehta

3. SELF JOIN

Find pairs of employees who work in the **same department**.

SELECT

e1.Emp_Name AS Employee1,

e2.Emp_Name AS Employee2,

e1.Dept_ID

FROM Employee e1

JOIN Employee e2

ON e1.Dept_ID = e2.Dept_ID

AND e1.Emp_ID < e2.Emp_ID;

Employee1	Employee2	Dept_ID
Aarav Sharma	Riya Gupta	1
Aarav Sharma	Arjun Verma	1
Aarav Sharma	Ananya Singh	1
Aarav Sharma	Karan Mehta	1
Aarav Sharma	Priya Sharma	1
Riya Gupta	Arjun Verma	1
Riya Gupta	Ananya Singh	1
Riya Gupta	Karan Mehta	1
Riya Gupta	Priya Sharma	1
Arjun Verma	Ananya Singh	1
Arjun Verma	Karan Mehta	1
Arjun Verma	Priya Sharma	1
Ananya Singh	Karan Mehta	1
Ananya Singh	Priya Sharma	1
Karan Mehta	Priya Sharma	1
Neha Kapoor	Rahul Jain	2
Neha Kapoor	Simran Kaur	2
Neha Kapoor	Aditya Roy	2
Neha Kapoor	Megha Joshi	2
Neha Kapoor	Vivek Malhotra	2
Rahul Jain	Simran Kaur	2
Rahul Jain	Aditya Roy	2
Rahul Jain	Megha Joshi	2
Rahul Jain	Vivek Malhotra	2
Simran Kaur	Aditya Roy	2
Simran Kaur	Megha Joshi	2
Simran Kaur	Vivek Malhotra	2
Aditya Roy	Megha Joshi	2
Aditya Roy	Vivek Malhotra	2
Megha Joshi	Vivek Malhotra	2
Ishita Rao	Manish Kumar	3
Ishita Rao	Pooja Agarwal	3
Ishita Rao	Rohit Bansal	3
Ishita Rao	Kavya Nair	3
Ishita Rao	Sahil Gupta	3
Manish Kumar	Pooja Agarwal	3
Manish Kumar	Rohit Bansal	3
Manish Kumar	Kavya Nair	3
Manish Kumar	Sahil Gupta	3
Pooja Agarwal	Rohit Bansal	3
Pooja Agarwal	Kavya Nair	3
Pooja Agarwal	Sahil Gupta	3
Rohit Bansal	Kavya Nair	3
Rohit Bansal	Sahil Gupta	3

Kavya Nair	Sahil Gupta	3
Tanya Singh	Yash Verma	4
Tanya Singh	Nisha Patel	4
Tanya Singh	Aman Khan	4
Tanya Singh	Shreya Das	4
Tanya Singh	Dev Sharma	4
Yash Verma	Nisha Patel	4
Yash Verma	Aman Khan	4
Yash Verma	Shreya Das	4
Yash Verma	Dev Sharma	4
Nisha Patel	Aman Khan	4
Nisha Patel	Shreya Das	4
Nisha Patel	Dev Sharma	4
Aman Khan	Shreya Das	4
Aman Khan	Dev Sharma	4
Shreya Das	Dev Sharma	4
Akash Yadav	Sneha Roy	5
Akash Yadav	Varun Singh	5
Akash Yadav	Komal Gupta	5
Akash Yadav	Nitin Sharma	5
Akash Yadav	Aditi Mehta	5
Sneha Roy	Varun Singh	5
Sneha Roy	Komal Gupta	5
Sneha Roy	Nitin Sharma	5
Sneha Roy	Aditi Mehta	5
Varun Singh	Komal Gupta	5
Varun Singh	Nitin Sharma	5
Varun Singh	Aditi Mehta	5
Komal Gupta	Nitin Sharma	5
Komal Gupta	Aditi Mehta	5
Nitin Sharma	Aditi Mehta	5

4. Three-Way JOIN

Display employee name, department name, and project name.

SELECT

e.Emp_Name, d.Dept_Name, p.Project_Name, e.Salary

FROM Employee e

JOIN Department d

ON e.Dept_ID = d.Dept_ID

JOIN Project p

ON e.Project_ID = p.Project_ID;

Emp_Name	Dept_Name	Project_Name	Salary
Aarav Sharma	IT	Website Development	55000.00
Riya Gupta	IT	Mobile Application	62000.00
Arjun Verma	IT	Website Development	48000.00
Ananya Singh	IT	Mobile Application	70000.00
Karan Mehta	IT	Website Development	45000.00
Priya Sharma	IT	Mobile Application	58000.00
Neha Kapoor	HR	Recruitment System	50000.00
Rahul Jain	HR	Recruitment System	65000.00
Simran Kaur	HR	Recruitment System	42000.00
Aditya Roy	HR	Recruitment System	46000.00
Megha Joshi	HR	Recruitment System	55000.00
Vivek Malhotra	HR	Recruitment System	72000.00
Ishita Rao	Finance	Financial Analysis	60000.00
Manish Kumar	Finance	Financial Analysis	75000.00

Pooja Agarwal	Finance	Financial Analysis	52000.00
Rohit Bansal	Finance	Financial Analysis	68000.00
Kavya Nair	Finance	Financial Analysis	58000.00
Sahil Gupta	Finance	Financial Analysis	80000.00
Tanya Singh	Marketing	Marketing Campaign	47000.00
Yash Verma	Marketing	Marketing Campaign	56000.00
Nisha Patel	Marketing	Customer Management	62000.00
Aman Khan	Marketing	Customer Management	44000.00
Shreya Das	Marketing	Marketing Campaign	51000.00
Dev Sharma	Marketing	Customer Management	59000.00
Akash Yadav	Operations	Inventory System	49000.00
Sneha Roy	Operations	Inventory System	57000.00
Varun Singh	Operations	Business Automation	63000.00
Komal Gupta	Operations	Inventory System	46000.00
Nitin Sharma	Operations	Business Automation	54000.00
Aditi Mehta	Operations	Business Automation	69000.00

5. Correlated Subquery

Find employees whose salary is greater than the **average salary of their own department**.

SELECT

e.Emp_ID, e.Emp_Name, e.Dept_ID, e.Salary

FROM Employee e

WHERE e.Salary >

(

SELECT AVG(e2.Salary)

FROM Employee e2

WHERE e2.Dept_ID = e.Dept_ID

);

Emp_ID	Emp_Name	Dept_ID	Salary
2	Riya Gupta	1	62000.00
4	Ananya Singh	1	70000.00
6	Priya Sharma	1	58000.00
8	Rahul Jain	2	65000.00
12	Vivek Malhotra	2	72000.00
14	Manish Kumar	3	75000.00
16	Rohit Bansal	3	68000.00
18	Sahil Gupta	3	80000.00
20	Yash Verma	4	56000.00
21	Nisha Patel	4	62000.00
24	Dev Sharma	4	59000.00
26	Sneha Roy	5	57000.00
27	Varun Singh	5	63000.00
30	Aditi Mehta	5	69000.00

6. EXISTS

Find departments that have at least one employee earning more than ₹70,000.

SELECT

d.Dept_ID,

d.Dept_Name

FROM Department d

WHERE EXISTS

(

SELECT 1

FROM Employee e

WHERE e.Dept_ID = d.Dept_ID

AND e.Salary > 70000

);

Dept_ID	Dept_Name
2	HR
3	Finance

7. EXISTS with Projects

Find employees who are assigned to a project.

SELECT

e.Emp_ID,

e.Emp_Name

FROM Employee e

WHERE EXISTS

(

SELECT 1

FROM Project p

WHERE p.Project_ID = e.Project_ID

);

Emp_ID	Emp_Name
1	Aarav Sharma
3	Arjun Verma
5	Karan Mehta
2	Riya Gupta
4	Ananya Singh
6	Priya Sharma
7	Neha Kapoor
8	Rahul Jain
9	Simran Kaur
10	Aditya Roy
11	Megha Joshi
12	Vivek Malhotra
13	Ishita Rao
14	Manish Kumar
15	Pooja Agarwal
16	Rohit Bansal
17	Kavya Nair
18	Sahil Gupta
19	Tanya Singh
20	Yash Verma
23	Shreya Das
21	Nisha Patel
22	Aman Khan
24	Dev Sharma
25	Akash Yadav
26	Sneha Roy
28	Komal Gupta
27	Varun Singh
29	Nitin Sharma
30	Aditi Mehta

8. Simulated INTERSECT

MySQL does not traditionally use the INTERSECT operator in the same way as some other SQL systems, so we can simulate it using INNER JOIN.

Find employees who belong to the IT department AND have salary greater than ₹60,000

```
SELECT Emp_ID, Emp_Name
```

```
FROM Employee
```

```
WHERE Dept_ID = 1
```

```
AND Emp_ID IN
```

```
(
  SELECT Emp_ID
  FROM Employee
  WHERE Salary > 60000
);
```

Emp_ID	Emp_Name
2	Riya Gupta
4	Ananya Singh

9. Simulated EXCEPT

Suppose we want employees who are **not working in the IT department**.

```
SELECT Emp_ID, Emp_Name
```

```
FROM Employee
```

```
WHERE Dept_ID <> 1;
```

Emp_ID	Emp_Name
7	Neha Kapoor
8	Rahul Jain
9	Simran Kaur
10	Aditya Roy
11	Megha Joshi
12	Vivek Malhotra
13	Ishita Rao
14	Manish Kumar
15	Pooja Agarwal
16	Rohit Bansal
17	Kavya Nair
18	Sahil Gupta
19	Tanya Singh
20	Yash Verma
21	Nisha Patel
22	Aman Khan
23	Shreya Das
24	Dev Sharma
25	Akash Yadav
26	Sneha Roy
27	Varun Singh
28	Komal Gupta
29	Nitin Sharma
30	Aditi Mehta

10. EXPLAIN – INNER JOIN

EXPLAIN shows how MySQL plans to execute a query.

EXPLAIN

SELECT

e.Emp_Name,

d.Dept_Name

FROM Employee e

INNER JOIN Department d

ON e.Dept_ID = d.Dept_ID;

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	Dept_ID	Dept_ID	5	db_4546kyfyj.d.Dept_ID	6	100	null

11. EXPLAIN – LEFT JOIN

EXPLAIN

SELECT

d.Dept_Name,

e.Emp_Name

FROM Department d

LEFT JOIN Employee e

ON d.Dept_ID = e.Dept_ID;

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	null	null	null	null	5	100	null
1	SIMPLE	e	null	ref	Dept_ID	Dept_ID	5	db_4546kyfyj.d.Dept_ID	6	100	null

12. EXPLAIN – Three-Way JOIN

EXPLAIN

SELECT

e.Emp_Name,

d.Dept_Name,

p.Project_Name

FROM Employee e

JOIN Department d

ON e.Dept_ID = d.Dept_ID

JOIN Project p

ON e.Project_ID = p.Project_ID;

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	Dept_ID,Project_ID	Dept_ID	5	db_4546kyfyj.d.Dept_ID	6	100	Using where
1	SIMPLE	p	null	eq_ref	PRIMARY	PRIMARY	4	db_4546kyfyj.e.Project_ID	1	100	null

13. EXPLAIN – Correlated Subquery

EXPLAIN

SELECT

e.Emp_Name,

e.Salary

FROM Employee e

WHERE e.Salary >

(

SELECT AVG(e2.Salary)

FROM Employee e2

WHERE e2.Dept_ID = e.Dept_ID

);

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	PRIMARY	e	null	ALL	null	null	null	null	30	100	Using where
2	DEPENDENT SUBQUERY	e2	null	ref	Dept_ID	Dept_ID	5	db_4546kyfyj.e.Dept_ID	6	100	null